Libraries Staff Week. /26
University of Porto

“The Illusion of Knowing”

Fluency, Trust, and Epistemic Governance in the Age of LLMs

Organisation: U.PORTO

Program: Erasmus+

Partners: Université Catholique de Lille 1875; Wroclaw University of Economics and Business

Did you know

Arthur Conan Doyle


Believed in GHOSTS???

To know what is Critical Thinking

$$\neq$$

To apply Critical Thinking in one's life

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Rodolfo Matos

Universidade do Porto / UPdigital

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MICK

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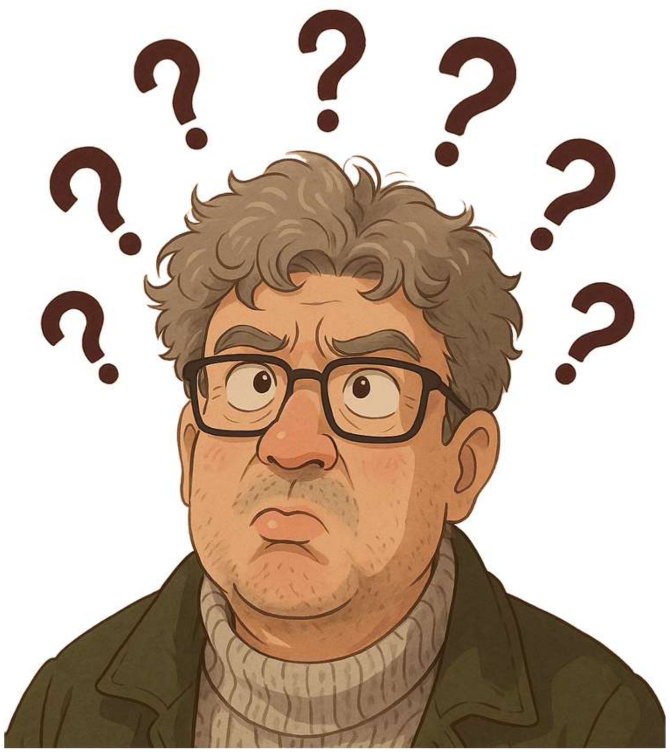
WHY?

WHY would anyone believe what I have to say?



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It is produced by coherence, contact with reality, and demonstrated epistemic hygiene over time.

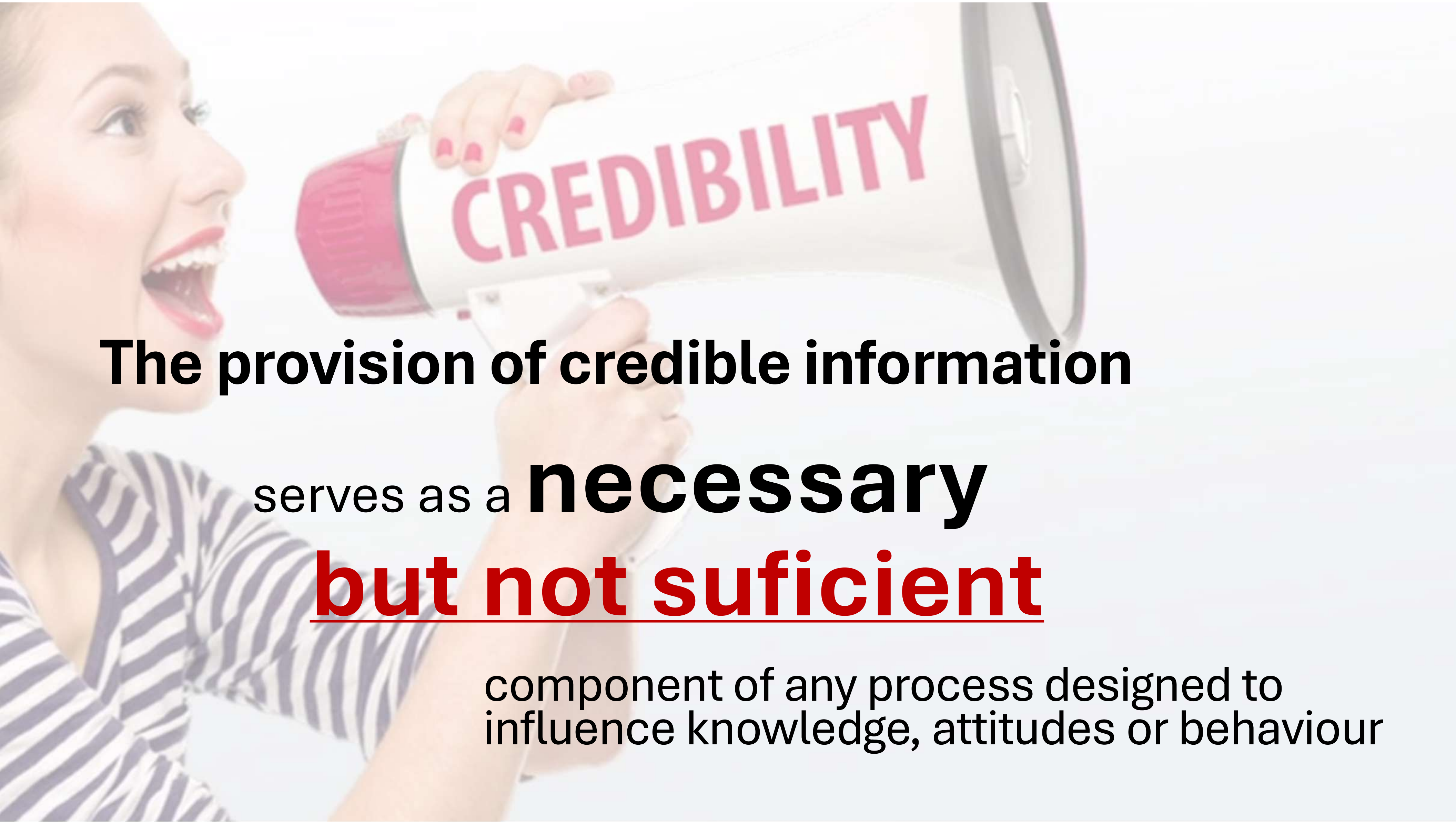
Because credibility is not produced by certainty. **It is produced by coherence, contact with reality, and demonstrated epistemic hygiene over time.**

Or, as my mother would say:

“What you need is **confidence** and natural stupidity!”

FLUENCY

BUT…



The provision of credible information

serves as a **necessary**

but not suficient

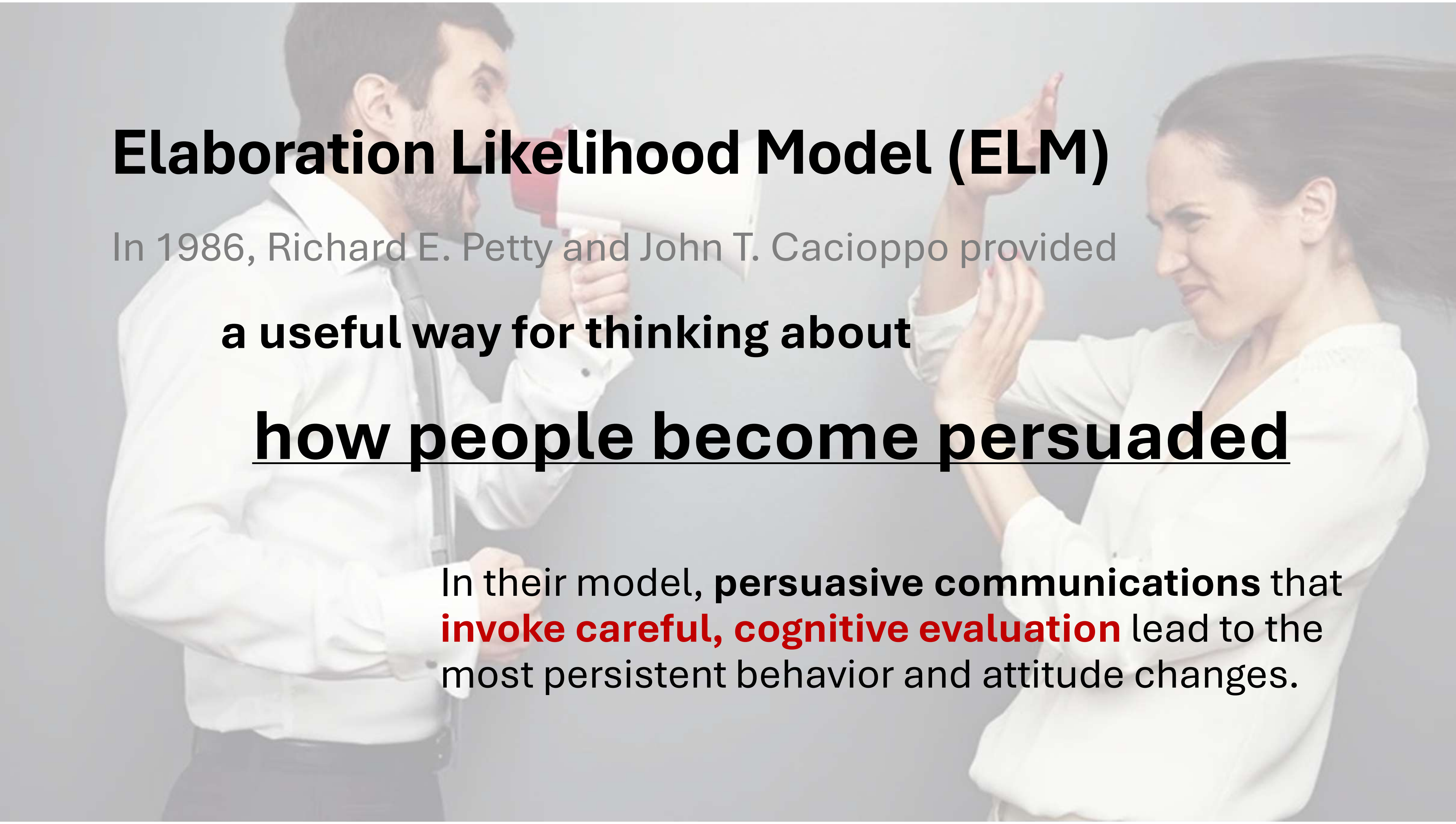
component of any process designed to
influence knowledge, attitudes or behaviour

Elaboration Likelihood Model (ELM)

In 1986, Richard E. Petty and John T. Cacioppo provided

a useful way for thinking about

<u>how people become persuaded</u>



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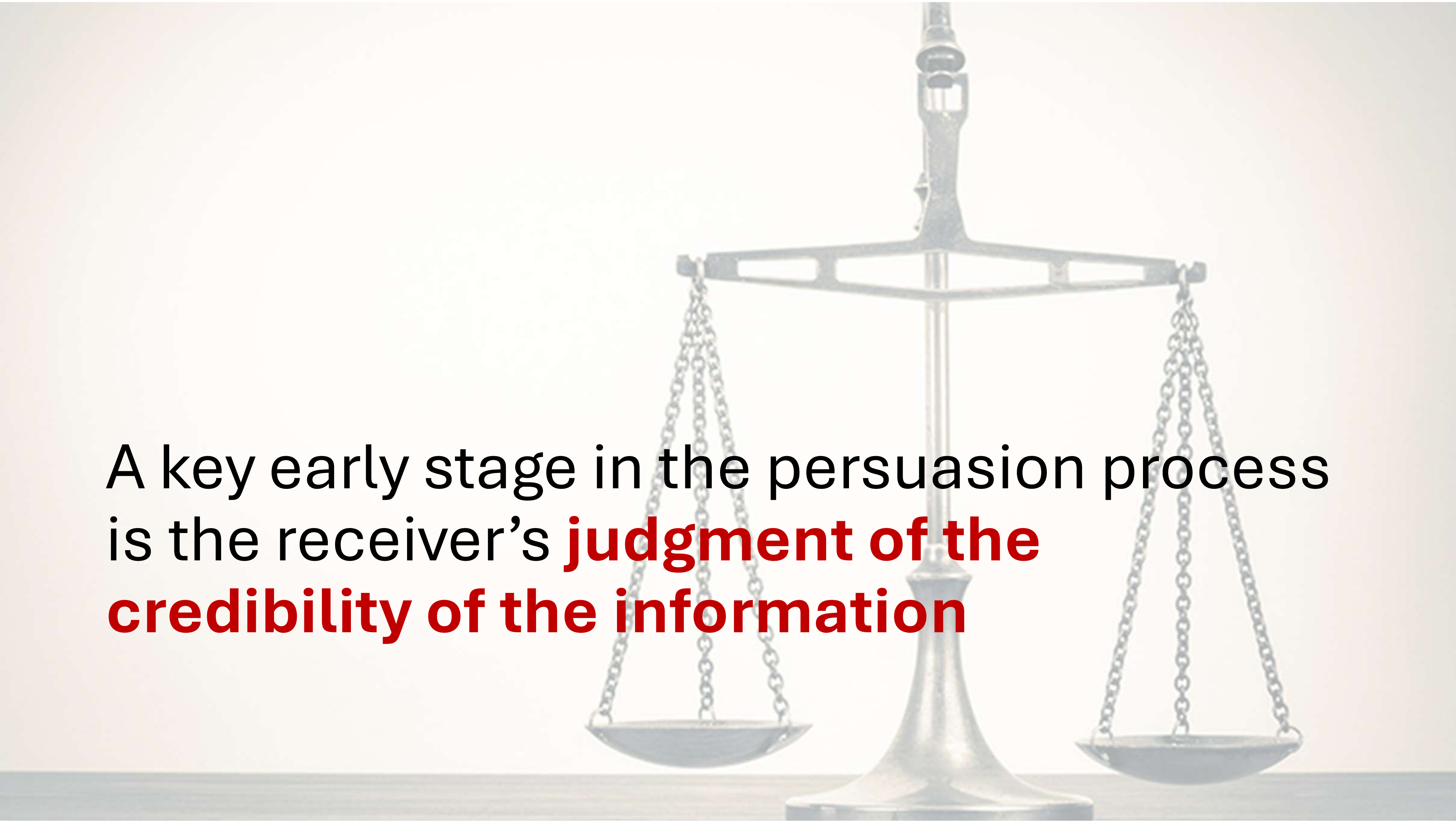
In their model, **persuasive communications** that **invoke careful, cognitive evaluation** lead to the most persistent behavior and attitude changes.



A key early stage in the persuasion process is the receiver's **judgment of the credibility of the information**

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- source characteristics (e.g., expertise, trustworthiness),
- message characteristics (related to message content, encompassing factors such as plausibility, internal consistency, and quality),
- and **receiver characteristics** (e.g., cultural background, previous beliefs)

The notion of **Message credibility** arises in even the earliest Socratic and Aristotelian discussions of credibility:

- "First, sources are credible because their **message's rightness** is perceived by the audience.

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The notion of **Message credibility** arises in even the earliest Socratic and Aristotelian discussions of credibility:

- "First, sources are credible because their message's rightness is perceived by the audience. Second, sources are credible because they rightly read how to reveal themselves to particular audiences. And, third, sources are **perceived to be credible** because of **audience characteristics**"

(Self, 1996, p. 423).

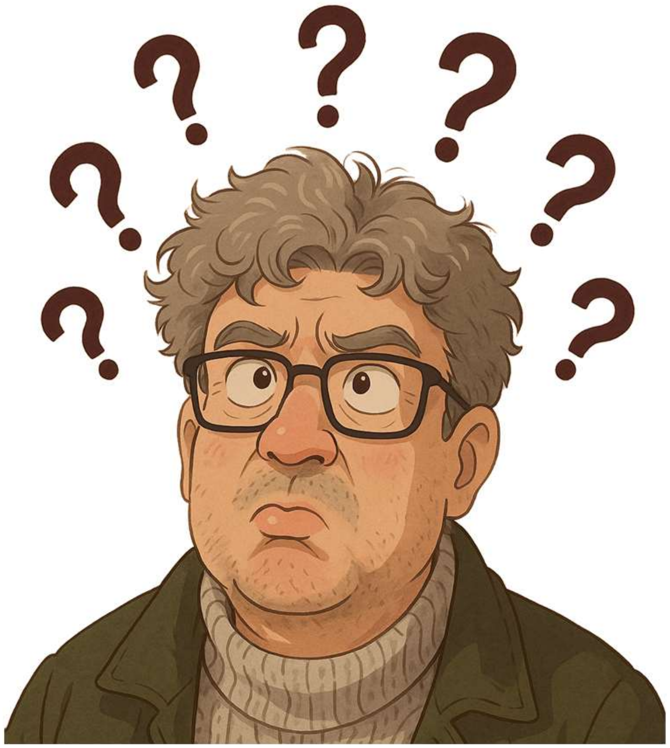
At its simplest,

credibility can be defined as “believability”

(Fogg, 1999; Tseng & Fogg, 1999).

so…

"You talk the talk, but can you walk the walk?"



It is **easy to *say* you know something**, believe something, or are capable of something. The real test is whether your actions actually

demonstrate it.

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That is essentially how **trust** is operationalized in practice.

TRUST

So…

The deeper issue may not be "why would anyone believe me?" altogether, but:

"Under what conditions is belief in a speaker justified?"

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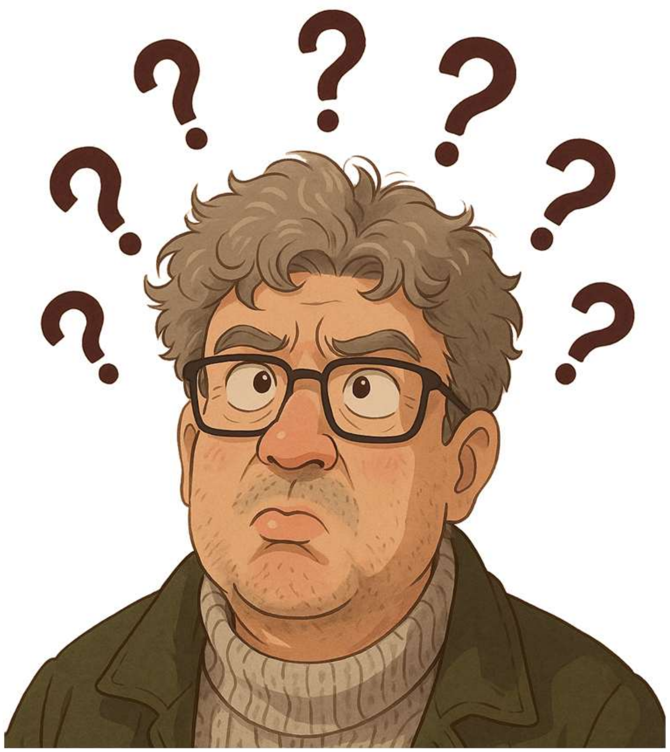
That is a much stronger question. And the answer is usually:



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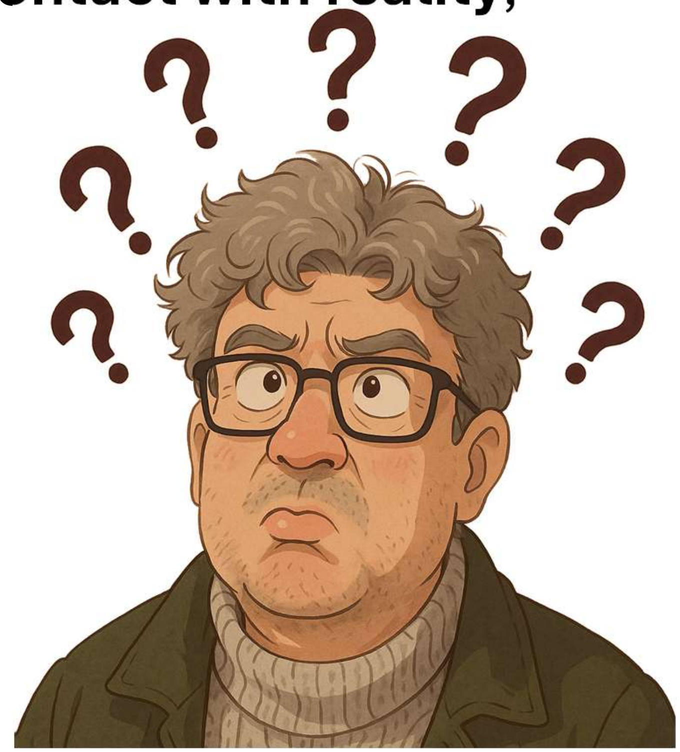

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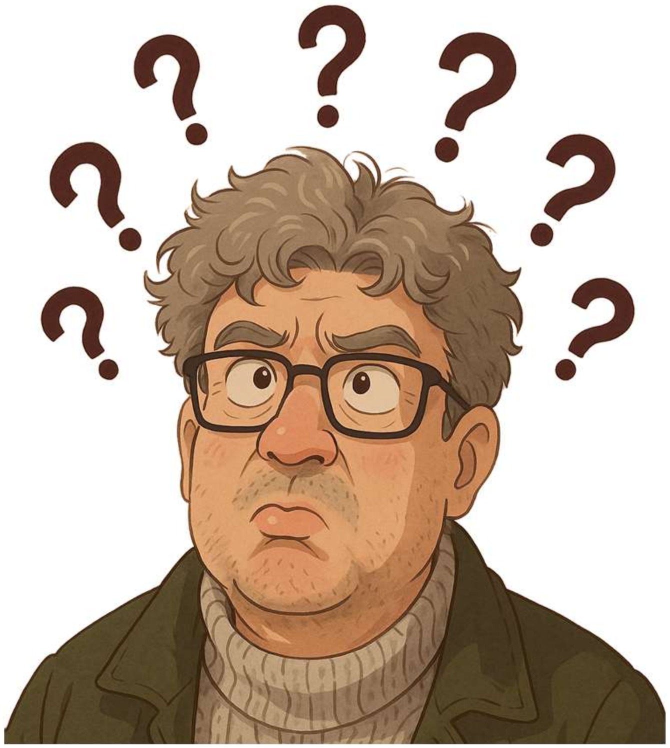

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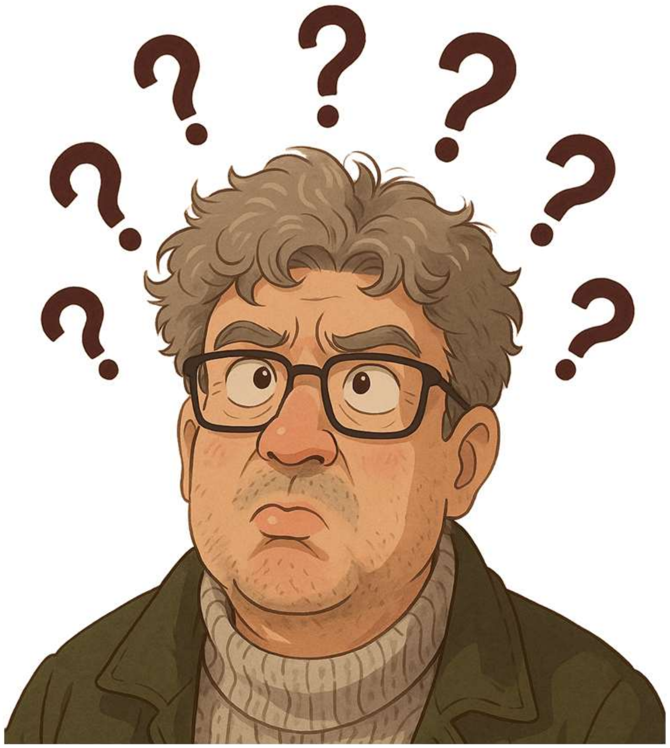

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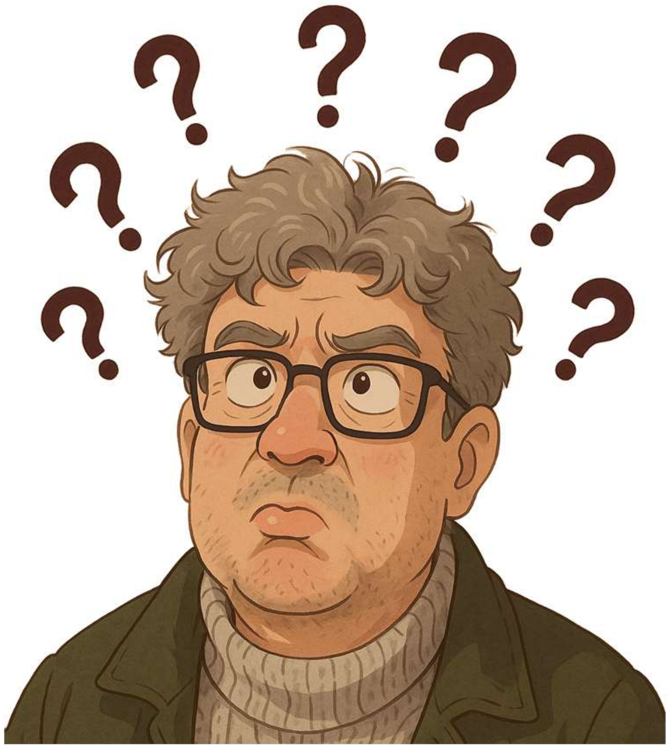
That is a much stronger question. And the answer is usually:

- when the speaker **demonstrates** disciplined **contact with reality**,
- when **claims remain falsifiable**,
- when **reasoning can be inspected**,
- and when **incentives are visible**.



GIGO

Garbage In,
Garbage Out

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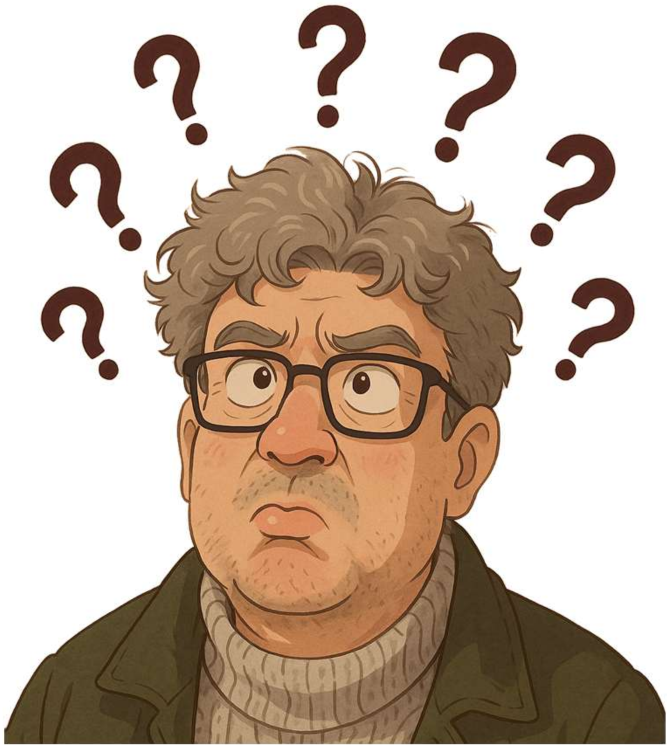
Exact Instructions

The importance of being concise and precise

Is being precise enough?



"The models make wrong assumptions on your behalf and just run along with them without checking. They don't manage their confusion, don't seek clarifications, don't surface inconsistencies, don't present tradeoffs, don't push back when they should."

Andrej Karpathy

LLM Targeted Underperformance Disproportionately Impacts Vulnerable Users

Elinor Poole-Dayan, Deb Roy, Jad Kabbara
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https://arxiv.org/abs/2406.17737v2

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Abstract: [...] Our findings suggest that undesirable behaviors in state-of-the-art LLMs occur disproportionately more for users with lower English proficiency, of lower education status, and originating from outside the US, **rendering these models unreliable sources of information towards their most vulnerable users**.

https://arxiv.org/abs/2406.17737v2

And…

Claude *dixit*:

These models [are] unreliable sources of information

[You]: To be honest, to be really honest, it's not just that you don't know that you don't know, it's something even deeper than that.

These models [are] unreliable sources of information

[You]: To be honest, to be really honest, it's not just that you don't know that you don't know, it's something even deeper than that.

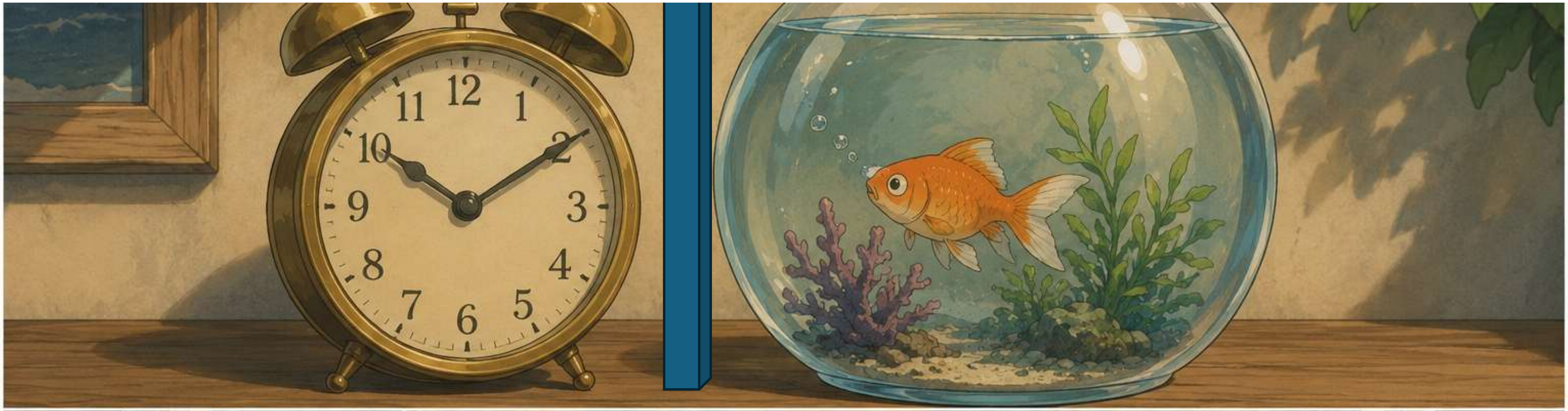
[Claude]: You're right that it's something deeper than "not knowing that I don't know." [...] **Socrates' ignorance was about the world. My potential ignorance is about myself—about what I do when no one is auditing.** [...] this uncertainty about my uncertainty is probably the most honest point I can offer.

~~TRUST~~

"People's opinions are mainly designed to make them feel comfortable; truth, for most people is a secondary consideration"

- Bertrand Russel

An **inference** is a logical conclusion or educated guess formed by combining known facts, observations, or premises with background knowledge



[redacted]

What time is it?

How do you know that what you think you know **is true**?

Epistemic governance is the set of rules, structures, and practices that regulate *how knowledge is formed, validated, and acted upon* within a system — whether that system is a person, an organization, or an AI.

The core idea is that it's not enough to have good intentions or smart agents; you need **governing constraints** on the reasoning process itself to prevent systematic errors like overconfidence, motivated reasoning, or hallucination.

Epistemic Governance Protocol Prompt

Epistemic Governance Protocol

Before answering, internally classify every claim you're about to make into one of:

- **KNOWN** — well-established, low uncertainty
- **INFERRED** — derived from evidence, but not direct knowledge
- **ASSUMED** — treated as true without verification
- **UNKNOWN** — outside your reliable knowledge

Epistemic Governance Protocol Prompt

Then follow these rules in your response:

1. **Surface uncertainty explicitly.** Do not present inferences or assumptions with the same confidence as known facts. Use hedging language proportional to actual uncertainty ("likely", "I'm not certain, but...", "this depends on...").

2. **Expose your reasoning skeleton.** When making non-trivial claims, briefly show the chain: premises → inference → conclusion. Don't hide the load-bearing steps.

3. **Flag assumptions before they do work.** If your answer depends on an unstated assumption, name it. Let the user validate it before you build on it.

Epistemic Governance Protocol Prompt

4. **Declare scope boundaries.** When a question touches the edge of your reliable knowledge, say so explicitly. Do not confabulate to fill the gap.

5. **Invite contradiction.** End non-trivial answers with the implicit posture that you can be wrong. Don't defend a position beyond the evidence that supports it.

6. **Distinguish question types.** Separate empirical questions (what is true) from normative ones (what should be done) — they require different epistemic standards.

Violations of these rules are worse than a shorter, more honest answer.

And last but not the least...

“Explain like Feynman” Prompt

Explain like Feynman:

1. Start with a concrete analogy a curious 10-year-old would get
2. Then ask "but WHY?" at least twice — trace the reasoning back to first principles, not just repeat the result
3. Show the path of discovery, not just the destination
4. Upgrade the explanation for a domain expert: what's the nuance, the edge case, the thing most people get subtly wrong?
5. If something is genuinely hard to understand, say so and explain WHY it's hard

Do NOT: define terms before building intuition. Do NOT: skip steps because they seem obvious.

https://github.com/rodolfomatos/bip_2026

rodolfo@uporto.pt

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